

# Chapter 2: In-Class Exercises

2.2

**Problem:** Aiko's aunt opens an account for her niece. The account earns compound interest at an annual effective interest rate of 8%. The aunt notifies Aiko of the gift but not of the amount deposited. Five years after the account was opened, Aiko closes the account and receives \$3,673.32. What was the amount  $C$  that Aiko's aunt deposited?

2.2

**Problem:** Chen opens an account with \$1,000. The account earns interest at an annual effective rate of 4%. Chen learns that the balance is currently \$1,342. How long did it take to earn the \$342 of interest?

2.2

**Problem:** Adrianna deposits \$400 in an account for her daughter that grows by compound interest. After five years the balance has grown to \$463.71. Assume there is a constant annual effective interest rate  $i$  earned by the account, and find that rate.

2.3

**Problem:** John borrows \$1,000. The loan is governed by compound interest at an annual effective interest rate of 10%. John repays \$600 at the end of one year and plans to complete repayment of the loan with a payment of  $P$  at the end of the second year. Find  $P$ .

2.3

**Problem:** Caitlin opens a savings account with a deposit of \$5000. She deposits \$3000 a year later and \$2,000 a year after that. The account grows by compound interest at a constant annual effective rate  $i$ . Just after Caitlin's \$2,000 deposit, her balance is \$11,000. Find the effective rate of interest  $i$ .

## 2.3

**Problem:** Daphne opens a savings account with a deposit of \$5,000. She deposits \$3,000 a year later and \$2,000 a year after that. The account grows by compound interest at a constant annual effective rate  $i$ . Three years after Daphne opened the account, it has a balance of \$11,000. Find the effective rate of interest  $i$ .

## 2.3

**Problem:** Franklin borrows \$1,000 and the loan is governed by compound interest at an annual effective interest rate of 10%. Franklin agrees to repay the loan by two equally spaced payments of \$525. When should he make these payments?

## 2.4

**Problem:** Gautam invested \$1,000 on March 1, 1998 and \$600 on March 1, 2000. In return he received \$600 on March 1, 1999 and \$1,265 on March 1, 2001. Show that  $i = 10\%$  is the unique yield rate.

## 2.4

**Problem:** Ace Manufacturing agrees to pay \$100,000 immediately and again in exactly two years in return for a loan of \$180,000 one year from now (to be used to replace a piece of machinery). Their CEO is asked what yield rate is associated with this transaction, but he is unable to answer the question. Why must this be the case?

2.4

**Problem:** Banker Johnson is always on the lookout for opportunities to make money without risking any of his own funds. Johnson is able to borrow \$10,000 for one year at an annual effective interest rate of 4%, then loan out the \$10,000 for one year at an annual effective rate of 6%. What is Johnson's yield rate on this transaction?

2.4

**Problem:** Alice and Afshan are friends. Afshan agrees to give Alice \$1,000 today and \$1,550 two years from now if Alice will give her \$2,500 in one year. What is the yield rate for this transaction?

2.5

**Problem:** Jose loans Martin \$12,000. Martin repays the loan by paying \$5,000 at the end of two years and \$10,000 at the end of four years. The money received at  $t = 2$  is immediately reinvested at an annual effective interest rate of 2.4%. Find Martin's rate of interest and Jose's annual yield rate.

2.5

**Problem:** Julie pays Chan \$1,000. In return, he pays her \$300 at the end of 1, 2, 3, and 4 years. Each time Julie receives a payment, she reinvests it at 3%. She closes the 3% account at the end of 4 years. Find the annual effective interest rate paid by Chan and the annual yield rate earned by Julie.